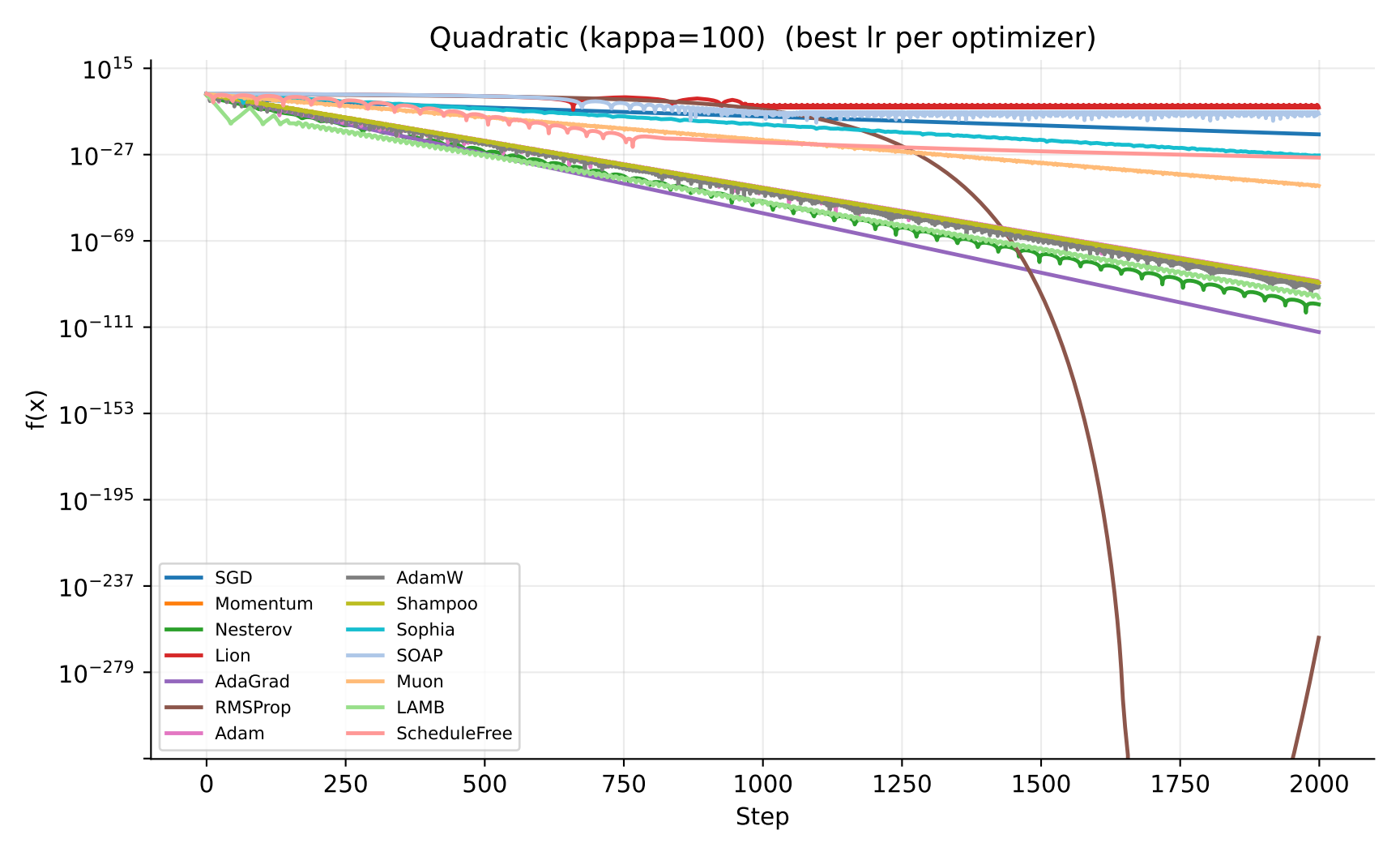

Quadratic (kappa=100) (best lr per optimizer)
f(x)
Step
10^{15}
10^{-27}
10^{-69}
10^{-111}
10^{-153}
10^{-195}
10^{-237}
10^{-279}
0
250
500
750
1000
1250
1500
1750
2000
SGD
Momentum
Nesterov
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
ScheduleFree